



Student Performance Prediction



Predicting Final Grades Using Machine Learning

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Week 4 Final Presentation

What Are We Predicting?

01 Target variable: G3 — Final Grade (scale 0–20)

02 Task type: Regression

03 Dataset: UCI Student Performance (Portuguese course)

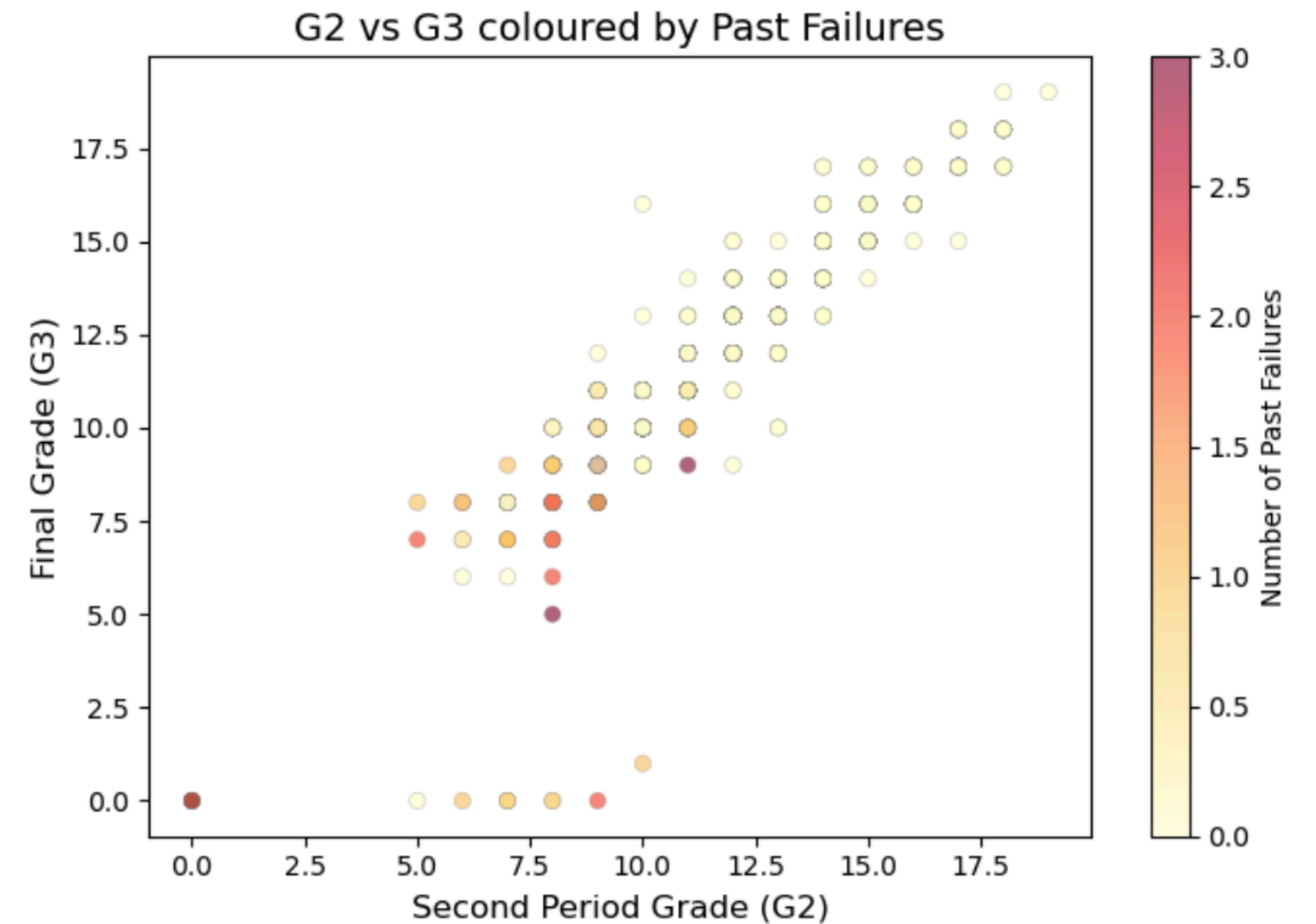
04 Size: 649 students, 33 features

05 Features include: prior grades (G1, G2), absences, study time, family background, and more

06 No missing values

Key Finding from Exploratory Analysis

Prior grades predict the final grade far more than any social or behavioural factor.



- G2 correlates with G3 at $r \approx 0.92$
- G1 correlates with G3 at $r \approx 0.83$
- Students with more past failures cluster at lower grades
- Absences showed almost no linear relationship ($r \approx -0.03$)

Models

Models used:

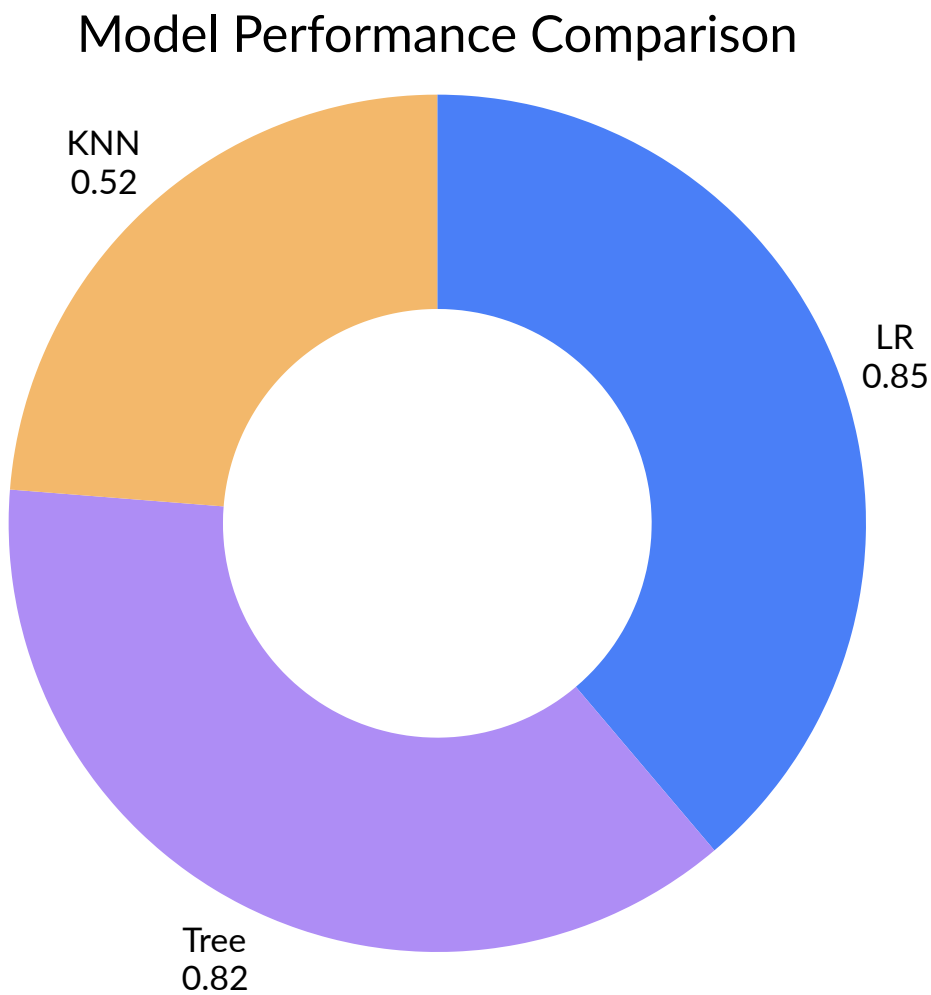
- 1) Linear Regression (Baseline)
- 2) K-Nearest Neighbors (KNN)
- 3) Decision Tree Regressor



Why these models?

- 1) strong linear relationships
- 2) KNN → captures local similarity
- 3) Tree → captures non-linear patterns

Results



► Best Model: Linear Regression

$R^2 = 0.85$
Lowest RMSE & MAE
Stable cross-validation performance

	Model	RMSE	MAE	R2	CV Mean (R2)	CV Std (R2)
0	Linear Regression (Baseline)	1.2149	0.7651	0.8487	0.8356	0.0433
1	KNN Regressor	2.1556	1.5569	0.5235	0.4901	0.0541
2	Decision Tree	1.3355	0.8011	0.8171	0.7618	0.0536

Problem Statement



- 01 **The "Zero-Grade" Mystery:** The model fails to predict a final grade of 0
- 02 **Specific Case:** Student with $G1=12$, $G2=10 \rightarrow$ Predicted $G3=10.5 \rightarrow$ Actual $G3=0$
- 03 **Root Cause:** A score of 0 represents a "Structural Zero" (dropout/absence), not academic performance
- 04 **Impact:** These outliers pull the regression line down, affecting overall accuracy.

Reflection & Future Steps

Me: *uses machine learning*

Machine: *learns*

Me:



- **Hypothesis Check:** Academic momentum (G1, G2) is the only dominant factor; behavioral traits were secondary.
- **Key Limitation:** Data Leakage — relying on G2 makes the model accurate but less useful for early-warning.
- **Future Solution:** A Two-Stage "Hurdle Model":
 1. Classifier to identify dropout risk.
 2. Regressor to predict the specific grade.

01



01. Academic Momentum Prior grades (G1/G2) are the only dominant predictors ($R^2 = 0.85$). Behavioral factors add minimal value once the academic track is set.

02



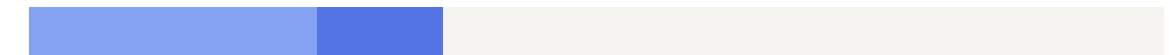
02. The "Zero" Gap The model fails to predict "Structural Zeros" (dropouts). These cases represent sudden behavioral shifts that linear regression cannot capture.

03



03. Future Strategy Transition to a two-stage "Hurdle Model" to separate dropout risk from grade estimation, creating a more robust early-warning system.

Conclusion



Thank You

